H-B-181D
February 9, 1976
SUPERSEDING
Fed. Spec. H-B-181C
March 21, 1968

FEDERAL SPECIFICATION

BRUSH, CHASSIS AND RUNNING GEAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type and size of chassis and running-gear cleaning brush (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951	-	Synthetic Filaments for Brushes; General Specification for
L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion)
PPP-B-1880	-	Brushes, Preparation for Delivery of

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 7920

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1363 - Measurement of Wood Moisture Content

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards

D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The brushes covered herein shall be a current standard product, except for any changes necessary to comply with specification requirements.

3.2 Materials. Materials shall conform to the applicable specifications and requirements specified hereinafter. Materials not specifically defined shall be of the quality used by the manufacturer in his commercial brushes.

3.2.1 Polypropylene filler material. The polypropylene filler material shall be 0.030 inches in diameter and shall conform to class P, type II of H-S-951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

3.2.2 Palmetto. The palmetto fiber shall be oil treated in accordance with commercial practice.

3.2.3 Wood. The block shall be hardwood of a dense close grained species such as beech, birch, sycamore, or maple. Prior to machining, the wood shall be dried to a moisture content not exceeding 12 percent when tested as specified in 4.2.1.1.

3.2.4 Staple wire. Staple wire shall be not less than 0.046 inch in diameter, commercial steel staple wire, corrosion-resistant or treated to resist corrosion.

3.2.5 Polypropylene molding material. The polypropylene molding material shall be virgin, general purpose homopolymer polypropylene meeting the requirements of L-P-394, except that the physical requirements after molding into a block shall be limited to those given in table I when tested as specified in 4.2.1.2. Clean, unburned plastic material in the form of imperfect parts, sprues, runners or other scrap of the same composition as virgin material and produced in the molding or finishing operations, may be reground and mixed with virgin material.

TABLE I. Requirements for polypropylene plastic

Property at 73°F (23°C)	Units	Property values
Density		
	Solid	gm/ml
	Foam	gm/ml
Tensile yield strength	psi	Min 4900
Izod impact strength	ft lb/in	Min 0.5
Rockwell hardness	R scale	Min 96

3.3 Construction. The brush shall conform to the requirements specified herein and in table II, and shall be similar to figure 1. Figure 1, is included for illustration only and is not intended to preclude the purchase of brushes otherwise in accordance with requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE II. Minimum physical requirements

Block dimensions, inches:	
Length:	
Overall	19-1/2
Brush part	3-1/8
Width:	
Handle (end)	3/4
Brush part	2-1/2
Thickness:	7/8
Holes:	
Diameter	9/32
Number	56
Angle (at sides and end)	30°
Filler material, as applicable:	
Weight, ounces: 1/	
Polypropylene	2-3/8
Oiled palmetto	3-7/8
Length clear of block, inches	2

1/ When tested as specified in 4.3.1, as applicable.

3.3.1 Filler material. The filler material shall be either polypropylene specified in 3.2.1 or palmetto fiber specified in 3.2.2.

3.3.2 Block. The block shall be fabricated from a single piece of the wood specified in 3.2.3 or the polypropylene specified in 3.2.5. Filler holes perimeters shall be spaced not less than 1/16 inch to an adjacent hole. The edges of the entire block shall be evenly rounded. A 3/16 diameter hole shall be centered 1-1/4 ± 1/4 inches from the end of the handle.

3.3.3 Set. The filler material shall be cut to double length and folded in the center and stapled into the holes of the block. The staples shall be made of the wire specified in 3.2.4. The tufts shall not loosen, nor pull out of the brush when tested as specified in 4.3.2.

3.4 Finish. The blocks shall be finished in accordance with the manufacturers commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

3.5 Identification marking. Each brush block shall be permanently and legibly indented or marked with the manufacturer's name or trade name, or trademark of such characters as to be identifiable with the manufacturer.

3.6 Workmanship. Blocks shall be free of loose grain, splinters and rough edges. Staples shall not be exposed. Filler material shall be clean, evenly trimmed and shall be free of foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Moisture content of wood blocks. When wood blocks are furnished, the moisture content of wood requirement as specified in 3.3.1 shall be determined in accordance with MIL-STD-1363 at time of assembly. The lot shall be all wood blocks of one type and size prepared under the same conditions, stored as a group, and submitted for inspection at one time. The sample unit shall be one block. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units. Three determinations shall be made on each sample unit. The average of the three determinations shall be the moisture content of the sample unit. Any sample unit of wood having more than 12 percent moisture content shall be classified as a defect.

4.2.1.2 Properties of plastic material. When plastic blocks are furnished, testing of the plastic blocks for density, tensile yield strength and Izod impact strength shall be in accordance with L-P-394. Testing of the plastic block for Rockwell Hardness shall be in accordance with procedure A of ASTM D785. Failure to comply shall be cause for rejection of the lot in process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

4.2.2 End product inspection. A lot shall consist of all brushes offered for inspection at one time. The sample unit for this inspection shall be one complete brush.

4.2.2.1 Visual examination. Examination shall be made of the brush for defects set forth in table III. The inspection level shall be level I with an AQL of 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE III. Visual examination

Examine	Defect
Finish	Not finished as specified
Block	Not single piece of wood; number of holes for filler material less than specified Edges not rounded; handle not provided with a hole: loose grain; splinter or rough edge
Staple	Exposed
Filler material	Not clean; not evenly trimmed; evidence of foreign material Palmetto fiber not oil treated
Marking	Omitted, incomplete, or illegible

4.2.2.2 Dimensional examination. Examination shall be made of the brush for defects in dimensions. Any dimension not within specification requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. Each sample brush shall be tested in accordance with 4.3.1.1 or 4.3.1.2 as applicable and 4.3.2. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

4.2.2.4 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected in accordance with the quality assurance provisions in PPP-B-1880.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

4.3 Tests.

4.3.1 Weight of filler material. The filler material shall be removed from the brush prior to weighing.

4.3.1.1 Polypropylene. Unconditioned polypropylene shall be weighed to determine compliance with table II.

4.3.1.2 Palmetto fiber. The palmetto fiber shall be conditioned for not less than 16 hours in air at a relative humidity of 50 ± 4 percent, and at a temperature of 73° + 2°F. The conditioned fiber shall then be weighed in the same atmosphere to determine compliance with table II.

4.3.2 Set retention. Grasp the tuft with a suitable holding device and apply a force by using a scale or weight of not less than 10 pounds in the direction tending to pull the tuft from the block to determine compliance with 3.3.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for level A, B, or C shall be in accordance with PPP-B-1880 (see 6.2).

6. NOTES

6.1 Intended use. The brush is intended to be used in cleaning of vehicular chassis and running gear.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of packaging and packing (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

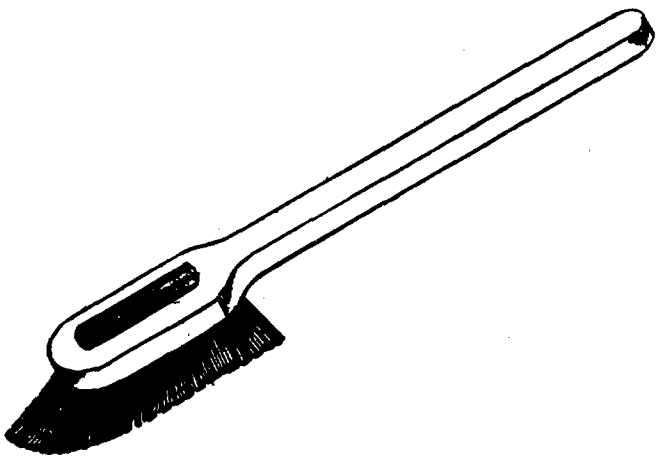


FIGURE I, BRUSH, CHASSIS AND RUNNING GEAR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

User activity:

Navy - MC

Civil Agency Coordinating Activity:

GSA - FSS

Preparing activity:

Army - GL

Project No. 7920-0201

Orders for this publication are to be placed with General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.